International Journal of Forecasting 41 (2025) 251–269

Contents lists available at ScienceDirect

International Journal of Forecasting

journal homepage: www.elsevier.com/locate/ijforecast

Dynamic time series modelling and forecasting of COVID-19 in Norway

Gunnar Bårdsen [a,b], Ragnar Nymoen [b,*]

[a] Norwegian University of Science and Technology, Norway
[b] University of Oslo, Norway

ARTICLE INFO

Keywords:
COVID-19 forecasting model
Dynamic time series modelling
Forecasting COVID-19 in Norway
New cases and hospital beds
Smooth transition regression
Forecasting policy effects

ABSTRACT

A framework for forecasting new COVID-19 cases jointly with hospital admissions and hospital beds with COVID-19 cases is presented. This project, dubbed CovidMod, produced 21 days ahead forecasts each working day from March 2021 to April 2022. Comparison of RMSFEs from that period, with the RMSFEs of the Norwegian Institute of Public Health (NIPH), favours the CovidMod forecasts, both for new cases and for hospital beds. Another comparison, with the short term forecasts produced by the Cardt method, shows little difference. Next, we present a new model where smooth transition regression is used as a feasible method to include forecasted effects of non-linear policy responses to the deviation between hospital beds and hospital bed capacity, on the forecasts of the original three variables. The forecasting performance of the model with endogenous policy effects is demonstrated retrospectively. It is suggested as a complementary approach to follow when the forecasted variables are generated from processes that include policy responses as realistic features.

© 2024 The Author(s). Published by Elsevier B.V. on behalf of International Institute of Forecasters. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

1. Introduction

During the COVID-19 pandemic, there was a surge in both demand for and supply of forecasts of how the virus was spreading in societies. Forecasts of new cases, and of new admissions to hospitals, were sometimes presented as premises for the many decisions that governments took to manage the crisis, in particular with regard to lock-downs and other non-pharmaceutical policy measures.

In Norway, the National Institute of Public Health (NIPH) published weekly *Situational awareness and forecasting for Norway* reports, with forecasts of the daily number of new infections and hospital beds. The main sequence of forecasts had a 21 days horizon and were based on the institute's National Regional Model.[1] One key parameter was the reproduction number (R) which was calibrated so that the model was adapted to historical hospital admissions and test data on a national level.[2]

The modelling group at NIPH revised reproduction numbers, in order to be able to capture new developments. Nevertheless there was never an expectation that the NIPH's 21 days forecast materialised unless the main drivers, e.g., virus characteristics and infection control measures, remained as they were close to the date of origin of the forecast over the forecast horizon.

Changing parameters in the underlying processes is a common problem in forecasting. A given model might work well within sample (the "training data set"), but yet produce poor forecasts if the structure of the underlying processes change in a time period after the preparation of the forecast. When forecasting pandemic data, being the result of many changing factors, structural breaks occurring after the forecast preparation date represent a potential problem.

* Corresponding author.
E-mail address: ragnar.nymoen@econ.uio.no (R. Nymoen).

[1] https://www.fhi.no/en/id/infectious-diseases/coronavirus/coronavirus-modelling-at-the-niph-fhi/.

[2] As well as county level, but we only consider national forecasts in this paper.

https://doi.org/10.1016/j.ijforecast.2024.05.004
0169-2070/© 2024 The Author(s). Published by Elsevier B.V. on behalf of International Institute of Forecasters. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Ideal samples for estimation of the epidemiological *R*-number may be data from the early period of the pandemic, where the data were not so affected by lockdowns or other non-pharmaceutical policies and interventions, (Korolev, 2021). Yet, that estimated *R*-number would be of limited value in forecasting the pandemic after the intervention was started.

Even if all model-based forecasts are vulnerable to structural breaks and shocks that occur after the origin of the forecast, there might be strategies to derive forecasting models with a higher degree of robustness against breaks in relationships and parameters. Such approaches, which for the pandemic data may be complementary to epidemiological structural models, include trend models, stochastic trend models, autoregressive models, exponential-smoothing algorithms, smooth-transition and robust forecasting methods, see Li and Linton (2021), Harvey and Kattuman (2021), Doornik et al. (2022), Mills (2022), Fayomi et al. (2022) amongst others.

Although an epidemiological model, like the NIPH's model, is a good model for understanding the deeper processes of the pandemic, it can still happen that the model's forecasts are less accurate than a time series oriented empirical model. The choice of model providing the most accurate forecast is an empirical question—depending upon the changing pandemic characteristics—and is likely to change. If the accuracy (quality) of the forecasts of the pandemic is at all of interest, it follows that the forecast errors of alternative models should be analysed and compared with improved forecast performance in mind.

This paper contains results from a project that was based on the idea that new cases, hospital admissions, and the number of patients in hospital with COVID-19, can be modelled as an interrelated dynamic system — an approach not dissimilar to one often taken in macroeconometrics. Among the many statistical forecasting tools that have been used to forecast COVID-19 around the world, our approach has most in common with the branch that used difference equations, with or without unit-root restrictions. More specifically, the model belongs to the class of autoregressive systems. Within this approach, the reproduction number *R* becomes not a parameter, but a time-varying variable within a model forecasting new cases and hospitalisations. Hence, although our modelling approach was different from the epidemiological model, there is still some common ground in the form of R numbers.

In the next section we show that an important parameter of the forecasting model was the long-run root of the characteristic equation of the process assigned to the development of new cases. A value larger than one meant that the spread of the virus had become explosive, or "critical". The parameter was estimated together with the other parameters of the model and was a main driver of the forecasts. One hypothesis to investigate during the forecasting project was that non-pharmaceutical policy responses as well as vaccinations and virus mutations could lead to changes in the critical parameter, and as we show in the empirical sections of the paper, we were able to incorporate such effects in the forecasting models as it developed through practical use.

After next presenting the conceptual framework, we explain how it was made operational in the form of a multiple-equation time series model, and used for forecasting in the period from March 2021 and May 2022. One thing we note, is that the adaptation of the model structure to incorporate effects of policy measures, behavioural changes among the public etc, has been essential in order to produce forecasts with information value.

We also review comparisons of the accuracy of real time forecasts with NIPH-forecasts as well as with the COVID-19 forecasts, as part a multi-country project in short-term forecasting, produced by a group of researchers at the University of Oxford.

While we followed a practical and adaptive modelling strategy when we prepared real time forecasting, it is interesting to use the data as a training data set for a forecasting model which might be more robust to policy changes. Specifically, the last part of the paper introduces a new and extended version of the model framework, where there is a distinction between exogenous shocks/breaks, due to infections, and the effects of policy responses which are endogenous. That model, which is non-linear in parameters, is in principle less dependent on the assumption about existing policies continuing into the forecast period, (Hyndman, 2023). The paper shows the specification and estimation of a policy-change-responsive model based on Smooth Transition Regression (STR) — see Teräsvirta et al. (2010, Ch. 3.4) for background. We also give examples of ex-post forecasts, which indicate that this type of model might be a useful complementary addition to the toolbox.

2. A model based on autoregressive processes

The model belongs to the class of autoregressive models. If we let Y_t denote new cases on day t, the starting point may be taken to be the dynamic model equation:

$$Y_t = \alpha_0 + \sum_{i=1}^{p} \alpha_i Y_{t-i} + \gamma X_t + \epsilon_t, \tag{1}$$

where α_i, $i = 0, 1., \ldots, p$ and γ are parameters. X_t represents conditioning variables, which is interpreted broadly to include deterministic terms. Lags of X_t have been kept out the notation for simplicity. ϵ_t is an error term which we assume linearly independent of $Y_{t-1}, \ldots, Y_{t-1-p}$ and X_t.

Given the complexity of the true process that have generated $Y_t, t = 1, 2, \ldots$, it is plausible that the error-term process of the simple linear form of the explanatory part of the model (1) is non-stationary, with time dependent moments, although not integrated. Hence, right from the start it is clear that it will be near impossible to obtain a congruent model of the DGP. However, when the purpose is forecasting, it is possible that models which are non-congruent can perform relatively well, in particular in comparison with models with little, or very stylized dynamics. First and foremost, forecasting is a time-oriented activity, (Granger & Newbold, 1986).

Seasonality is a feature of the daily data for new cases. A time series model with a seasonal difference operator applied to the differenced series (ΔY_t), and with a

seasonal moving average term, was therefore a clear candidate, see e.g., Harvey (1993, Ch. 5.6) and Osborn (2002). However, in our project a point was made about letting the data speak about differencing (implying $k = 1$ in Eq. (2) below) or not. In addition, a simple autoregressive model was thought to be easiest to operate jointly with equations for hospital admissions and hospital beds (number of patients). Hence, it was decided to use models with $p = 7$ (or larger) in order to capture seasonality in a rough and ready way.

Eq. (1) can be re-written as:

$$Y_t = \alpha_0 + kY_{t-1} + \sum_{i=1}^{p-1} \alpha_i^{\dagger} \Delta Y_{t-i} + \gamma X_t + \epsilon_t, \tag{2}$$

where the coefficients $\alpha_i^{\dagger}$ $(i = 1, 2, \ldots, p-1)$ are combinations of the autoregressive coefficients and k is:

$$k = \sum_{i=1}^{p} \alpha_i.$$

We note that $k = 1$ implies:

$$\Delta Y_t = \alpha_0 + \sum_{i=1}^{p-1} \alpha_i^{\dagger} \Delta Y_{t-i} + \gamma X_t + \epsilon_t, \tag{3}$$

and hence that Y_t is a non-stationary time series. Moreover, since the relevant solutions of the stochastic difference equations are future-independent solutions, $k > 1$ will result in explosive solutions and $k < 1$ in stable solutions.

Hence, k is an important parameter. One could say that it is a critical parameter, since a value larger than one means that the number of new cases will grow exponentially. As long as COVID-19 created serious illness among large swaths of the population, one consequence could be that the health care system became overflowed with patients.

The parameter k can be estimated from the data almost in real time. Signs of changes in k can be investigated by the use of for example recursive or rolling regressions. It is possible to imagine k as a parameter that the health authorities would find useful to monitor, and as a parameter that they would want to push below 1 in order to check the explosive process and to secure that the health service system remained intact during the pandemic.

The estimates of k may change as a result of the evolutionary path taken by the virus, through the continuous development of new versions and through intermittent mutations. But as noted above, k may also change in response to non-pharmaceutical interventions (NPIs), and more gradually as a consequence of vaccination and of herd immunity.

If the autoregressive model is used to forecast new cases, feasible methods to identify and represent breaks in k (once they have occurred) need to be part of the forecasting practice and method. Of course, smaller and more gradual changes in k will be difficult to identify empirically. However, even getting the intermittent larger breaks in k into the forecasting model with little delay, may mean a difference between useful and useless forecasts during a pandemic. As noted, a main distinction can be drawn between autonomous breaks in k due to new virus versions and mutations, and breaks that come as a result of NPIs. In our real-time forecasting, the practical method was to adjust the estimation of k based on the evidence that could be gleaned from the data as it came in, both autonomous and policy induced.

In a section below we show an example of one specific (date late in November 2021) forecasting model. It illustrates how the estimated k parameter changed both gradually (captured by the model's vaccination variable), and had been found empirically to have changed abruptly during the estimation period of that particular model.

However, there are alternatives that could have improved on our informal methods. As one example of a way forward we show how effects of policy-induced structural breaks in k can be included in the forecasting model with the aid of non-linear (threshold) estimation.

Returning to Eq. (1), another way of re-writing it gives:

$$Y_t = \alpha_0 + \beta\left(\sum_{i=1}^{p-1} Y_{t-i}\right) + \sum_{i=1}^{p-1} \alpha_i^{\ddagger} \Delta Y_{t-i} + \gamma X_t + \epsilon_t, \tag{4}$$

where:

$$\beta = \frac{k}{p-1} \tag{5}$$

and $\alpha_i^{\ddagger}$ $(i = 1, 2, \ldots, p-1)$ are other combinations of the original autoregressive coefficients than the $\alpha_i^{\dagger}$s in (4).

The $\alpha_i^{\ddagger}$s can be expressed as:

$$\alpha_i^{\ddagger} = \frac{p-1-i}{p-1} \sum_{j=1}^{i} \alpha_j - \frac{i}{p-1} \sum_{j=i+1}^{p} \alpha_j,\ i = 1, 2, \ldots, p-1. \tag{6}$$

In the case of $p = 4$, the expressions become:

$$\alpha_1^{\ddagger} = \left[\frac{2}{3}\alpha_1 - \frac{1}{3}(\alpha_2 + \alpha_3 + \alpha_4)\right] = \left[\frac{2}{3}\alpha_1 + \frac{1}{3}\alpha_1^{\dagger}\right],$$
$$\alpha_2^{\ddagger} = \left[\frac{1}{3}(\alpha_1 + \alpha_2) - \frac{2}{3}(\alpha_3 + \alpha_4)\right] = \left[\frac{1}{3}\alpha_1 + \frac{2}{3}\alpha_2^{\dagger}\right],$$
$$\alpha_3^{\ddagger} = -\alpha_4 = \alpha_3^{\dagger}.$$

The accumulated number of confirmed cases for day t is denoted S_{Yt}. It is given by the definition:

$$S_{Yt} = Y_t + S_{Yt-1}. \tag{7}$$

The change in accumulated cases over a period of q days, $\Delta_q S_{Yt}$, is therefore the cumulative sum of new cases over q days:

$$\Delta_q S_{Yt} = S_{Yt} - S_{Yt-q} = \sum_{i=0}^{q-1} Y_{t-i} \tag{8}$$

and for $t-1$:

$$\Delta_q S_{Yt-1} = S_{Yt-1} - S_{Yt-1-q} = \sum_{i=0}^{q-1} Y_{t-1-i} = \sum_{i=1}^{q} Y_{t-i}. \tag{9}$$

If we set $q = p - 1$,

$$\begin{aligned}\Delta_{(p-1)}S_{Yt-1} &= S_{Yt-1} - S_{Yt-1-(p-1)} \\ &= S_{Yt-1} - S_{Yt-p} = \sum_{i=1}^{p-1} Y_{t-i}. \end{aligned} \tag{10}$$

Eq. (4) can therefore be expressed with the change $S_{Yt-1} - S_{Yt-p}$ on the right hand side:

$$Y_t = \alpha_0 + \beta(S_{Yt-1} - S_{Yt-p}) + \sum_{i=0}^{p-1} \alpha_i^{\ddagger} \Delta Y_{t-1-i} + \gamma X_t + \epsilon_t. \tag{11}$$

A reasonable working hypothesis is that the amount of virus in the population, call it S_t^* (which is not observable), is increasing in the accumulated number of confirmed cases over a period of p days:

$$S_t^* = f(S_{Yt-1} - S_{Yt-p}), f' > 0, \tag{12}$$

which shows that "behind" the simple autoregressive model for new cases, it is the "stock" of infectious virus that affects the flow of new cases. In that interpretation (11) can be said to bring forward the positive feedback between new cases and how much corona virus there is in the population.

Another reason for using (11) instead of (2) in the estimated forecasting model was more practical. The time series of new cases was by the specimen collection date. There was a 3–4 day delay between diagnosis and registration in the Norwegian Surveillance System for Communicable Diseases (MSIS). The number of cases was updated retroactively when new notifications arrived.[3] A consequence for forecasting was a "ragged edge problem": when we prepared a forecast for $T + 1$, $T + 2$...., $T + 21$ the correct numbers of cases for the days in the initialisation period, in particular Y_T, Y_{T-1} and Y_{T-2} were not observable.

We worked around this problem by substituting the preliminary data by "nowcasted" values $\hat{Y}_T$, $\hat{Y}_{T-1}$ and $\hat{Y}_{T-2}$. For this we made use of the number of cases ordered by the date of registration, which were not subject to revision, together with lags of the numbers of cases by the specimen collection dates. For this reason, the summation in (4) improved the 21 days ahead forecasts compared to (2), which would have been more vulnerable to inaccurate nowcasting of a particular single day, Y_T.

2.1. New cases, admissions to hospital and number of patients, a flow chart view

The model used for real-time forecasting includes four endogenous variables:

- Y_t, number of new confirmed cases with COVID-19 day t.
- $S_{Y,t}$, accumulated number of confirmed cases with COVID-19, day t.
- HA_t, number of new admissions to hospital with COVID-19, day t.
- HB_t, number of persons in hospital with COVID-19 day t. We follow the convention of Norwegian Institute of Publich Health and refer to this variable as *hospital beds*, although this term has the drawback that it can be interpreted as a capacity number (but perhaps not when used in the context of forecasting patients).

Fig. 1 shows a flow chart of the model, which was dubbed CovidMod. Early in the pandemic, new cases with COVID-19 were in practice imported, when people returned from locations where there already were outbreaks. This early development is indicated by the line from the *COVID-19 abroad* node in the flow chart. The "import channel" remained important during the whole pandemic, but it was checked and in periods practically cut off by forced and voluntary reductions in travel abroad and by the quarantine regime (indicated by the dotted line that goes from the node for *Mutations, NPIs and vaccination*).

We have not explained new cases as a direct result of international travel, and so such changes ended up in the error term in the model. However, once the number of new cases had become noticeable in Norway, the model could pick up the positive feedback loop between new cases and the number of persons with COVID-19 who could transmit the virus to other persons. This is indicated by the line with the double arrowheads between the $\Delta_q S_{Yt-1}$ node and Y_t node. This relationship, the pandemic feedback loop, underwent numerous changes during the period. Non-pharmaceutical measures (lockdowns in particular), meant that fewer new cases resulted from the existing $\Delta_q S_{Yt-1}$ in the population, while the lifting of those measures worked in the other direction. As the vaccination rate grew, less social distancing was necessary to control the pandemic.

Mutations of the virus played a major role. The feedback loop between $\Delta_q S_{Yt-1}$ and new cases was affected by which virus strain was dominant. This also played a role in determining how hospital admissions responded to more COVID-19 in the population, represented by the line between $\Delta_q S_{Yt-1}$ and HA_t. Eventually, the vaccination program became the single most important factor in breaking that relationship, so that the society could reopen without a dangerous increase in new hospitalisations.

In CovidMod, the number of hospital beds (HB) is determined by the law-of-motion relationship:

$$HB_t = HA_t - \delta HB_{t-1}, \tag{13}$$

where the "capital depreciation" parameter δ is estimated.

The system in the flowchart suggests multiple-equation modelling. In the CovidMod project sequential conditioning was used. The number of hospital beds was modelled conditional on hospital admissions which in turn were modelled by conditioning on new cases.

The main reason why we stopped short of modelling the number of deaths was practical. The number of deaths was not forecasted by the National institute of Public Health which provided the government with weekly forecasts, the main reference for our project. However, the task of modelling deaths would have been important on

[3] https://www.fhi.no/en/id/infectious-diseases/coronavirus/daily-reports/daily-reports-COVID19/

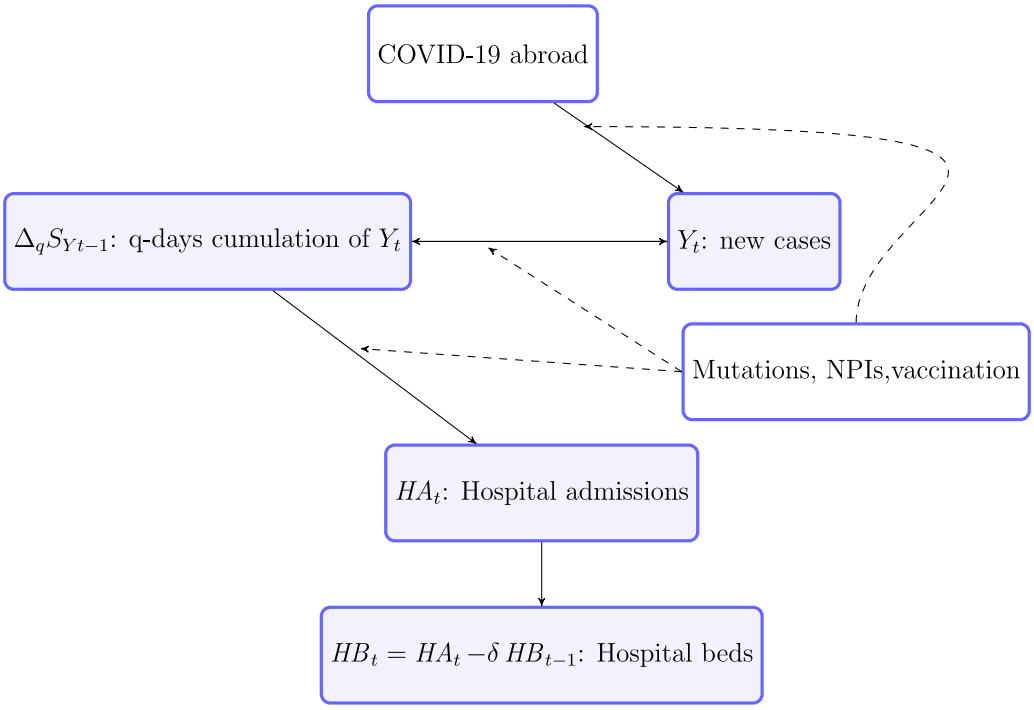


Fig. 1. Variables and relationships in CovidMod. Relationships that may change (structural breaks) as a consequence of virus mutations, NPIs (inducing behavioural changes) and vaccination, are indicated by dotted lines.

its own ground. Though very challenging, since it is plausible that the number of deaths depend on more factors than we included in the modelling work documented here, forecasting deaths should be included in a more complete model of the main pandemic indicators for Norway.

3. Specification, estimation and simulation of the model

As noted above, the detailed model specification was developed through practical forecasting in real time. A pilot model version was developed in the winter of 2020, soon after the outbreak of the pandemic. In the summer of 2020 the model became defunct, but when the forecasting was re-started in March 2021, it was possible to build on the pilot version from 2020. Then, from mid March 2021 and until early in May 2022, forecasts were produced five times weekly (i.e., work-days).

In our experience, it was important to keep an eye on forecast errors as they were revealed to us as forecasters. The errors could represent evidence that the incumbent model's assumptions were no longer valid, and that changes in the model were needed to avoid large forecast errors for the days ahead. Such changes could be to accommodate the effects of a new strain of the virus, or the effect of a policy response. Hence during the forecasting period, abrupt and more gradual changes in the dynamics of the pandemic were incorporated in the model by step-indicator variables (step dummies) and by variables that measured the percentage of the population that had received vaccination (one of more doses).

The step-indicator variables (step-dummies) can be divided into two categories:

- Abrupt exogenous events, like mutations (D).
- Non-pharmaceutical policy responses (G).

We next explain how variables of the two categories were included in the operative forecasting model. Since a new model was estimated for each of the forecasts, the project produced a whole sequel of models. As an example of the model specification in mid-November 2021, we consider the model estimated on the sample that begins with 21 February 2020 and ends with 22 November 2021.

The estimated equation for new cases Y_t, the empirical counterpart to Eq. (11), was:

$$\begin{aligned}\widehat{Y}_t &= \underset{(0.001)}{0.072}\,(S_{Yt-1} - S_{Yt-14}) \\ &+ \underset{(0.003)}{0.019}\,(S_{Yt-1} - S_{Yt-14})\,D_{\alpha t} \\ &+ \underset{(0.006)}{0.014}\,(S_{Yt-1} - S_{Yt-14})\,D_{\delta t} \\ &- \underset{(0.002)}{0.013}\,(S_{Yt-1} - S_{Yt-14})\,G_{Mar21t} \\ &+ \underset{(0.002)}{0.017}\,(S_{Yt-1} - S_{Yt-14})\,G_{Oct21t} \\ &- \underset{(0.011)}{0.051}\,(S_{Yt-1} - S_{Yt-14})\,VAC_t \\ &+ \sum_{i=0}^{13} \hat{\alpha}_i^{\ddagger} \Delta Y_{t-1-i} + \text{other short term factors.} \\ &\text{Estimation period: } 21.2.2020 - 22.11.2021,\ 641 \text{ obs.}\end{aligned} \tag{14}$$

The estimation method was OLS. Heteroscedasticity and autocorrelation consistent standard errors (HAC s.e.) are reported in parentheses below the estimated coefficients.

In this model equation, there is a reference estimate of β in the theoretical Eq. (11), namely 0.072 in the first row of (14).

In the second and third row, we find $D_{\alpha t}$ and $D_{\delta t}$, two step-dummies that are zero before the Alpha and Delta variants became dominant, and 1 after. They enter multiplicatively with $(S_{Yt-1} - S_{Yt-14})$. The positive coefficients imply that the positive feedback between new cases and the amount of transmittable corona disease became stronger when the two variants became dominant. When both dummies take the value 1, the implied estimate for β becomes 0.105 and the implied k-number becomes 1.36. Hence the process that generated new cases had become critical (explosive).

G_{Mar21t} represents the (renewed) lockdown in March 2021, and G_{Oct21t} is a dummy for a partial opening-up of the society in October 2021. The interpretation is that these changes in non-pharmaceutical policies worked as expected: The lockdown weakened the positive feedback loop, while the opening-up strengthened it.

The last of the multiplicative terms shows that a higher vaccination rate (VAC) had the expected dampening effect on the feedback loop, and can be imagined as a force that puts downward pressure on β and the critical parameter k as VAC_t increases in a continuous manner towards 1. However the estimated coefficient is quite large in magnitude, and a value of 0.4 implies an effect that is comparable the estimated effect of the NPI measures.

The second estimated equation in the model is the conditional model of new admissions to hospital with COVID-19 (HA):

$$\begin{aligned}\widehat{HA}_t &= \underset{(0.081)}{0.276} HA_{t-1} + \underset{(0.0014)}{0.0035} Y_t \\ &+ \underset{(0.00013)}{0.00039} (S_{Yt-3} - S_{Yt-19}) \\ &+ \underset{(0.0001)}{0.0004} (S_{Yt-3} - S_{Yt-19})\, D_{\alpha t} \\ &+ \underset{(0.0001)}{0.0005} (S_{Yt-3} - S_{Yt-19})\, G_{Oct21t} \\ &- \underset{(0.0002)}{0.0009} (S_{Yt-3} - S_{Yt-19})\, VAC_{t-7} \\ &+ \sum_{i=0}^{2} \hat{\gamma}_i \Delta HA_{t-1-i} + \text{other short term factors.}\end{aligned}$$

Estimation period: 14.7.2020 − 22.11.2021, 497 obs. (15)

The main explanatory factor is the number of new cases in the form of an intermediate term cumulative number $(S_{Yt-3} - S_{Yt-19})$ which is different from the one used in the Y-equation. The reason for not using the same virus stock variable in the equations was empirical and was based on statistical significance. However the difference is marginal. For example, the t-value of 3 in Eq. (15) is reduced to 2.5 if $(S_{Yt-3} - S_{Yt-19})$ is replaced by $(S_{Yt-1} - S_{Yt-14})$.

In addition there is an estimated impact effect of Y_t which at first can be difficult to rationalise, since there is normally a time lag between attracting the virus and being admitted to hospital. Although the statistical significance of Y_t was robust, there was a structural break at the end of April 2021. The estimated coefficient dropped, albeit not to zero. However, as there was no change in policies or in the virus at that point in time, we do not want to classify that regime shift as one or the other. Therefore, the reported coefficient of Y_t is the post-break estimate. Finally, as $(S_{Yt-1} - S_{Yt-14})$ is forecasted by the use of (14), keeping Y_t in the HA Eq. (16) did not represent a cost in the practical operation of the forecasting model.

Interestingly, the estimated model equation contained effects of virus mutation (but only of the Alpha variant), lock-down and vaccination, with the expected signs of the coefficients.

The third empirical equation in the model estimated on 22 November 2021 was the simple law of motion for the number of hospital beds with COVID-19:

$$\widehat{HB}_t = HA_t + \underset{(0.0029)}{0.898} HB_{t-1}.$$

Estimation period: 14.7.2020 − 22.11.2021, 497 obs. (16)

As noted, a realistic picture of the data generating process of new cases (Y) and hospital admissions (HA), is that it is complex and changing. In comparison the model equations are simple. Hence, there are several factors that are captured by the error terms of the estimated equations. As mentioned, import of virus through international travel has played a major role during the pandemic. This factor is not represented in CovidMod and is a source of simulation and forecast errors.

Fig. 2 shows how the model estimated on 22 November 2021 simulated the historical development of the variables from 1 July 2021 to 20 November 2021.

The dynamic simulation shown in Fig. 2 demonstrates that the model was able to explain many of the features of the data series.

It may also be of interest to look at this simulation through the lens of the time varying reproduction number, the R-number. Following Harvey and Kattuman (2021), it can be calculated as the ratio between the sum of new cases over the last four days and the corresponding sum for the previous four days. More generally:

$$R_{t,r,p} = \frac{\sum_{j=0}^{p-1} Y_{t-j}}{\sum_{j=r}^{p+r-1} Y_{t-j}}, \tag{17}$$

where the sum in the denominator starts at lag r and the sums of numerator and denominator can overlap.

Simulated R-numbers can be obtained by substituting actual numbers of new cases by simulated variable values. For the simulation of the number of new persons registered with COVID-19 (i.e., new cases) shown in Fig. 2, the time varying R-number using $p = r = 4$ is plotted in Fig. 3. The graph shows a simulated R-number close to one at the start of July 2021. It then increased gradually during July, the main summer holiday month which meant that many travelled abroad, although less than in normal times also. Towards the end of August, there was a marked spike in the simulated R-number. During September 2021 the

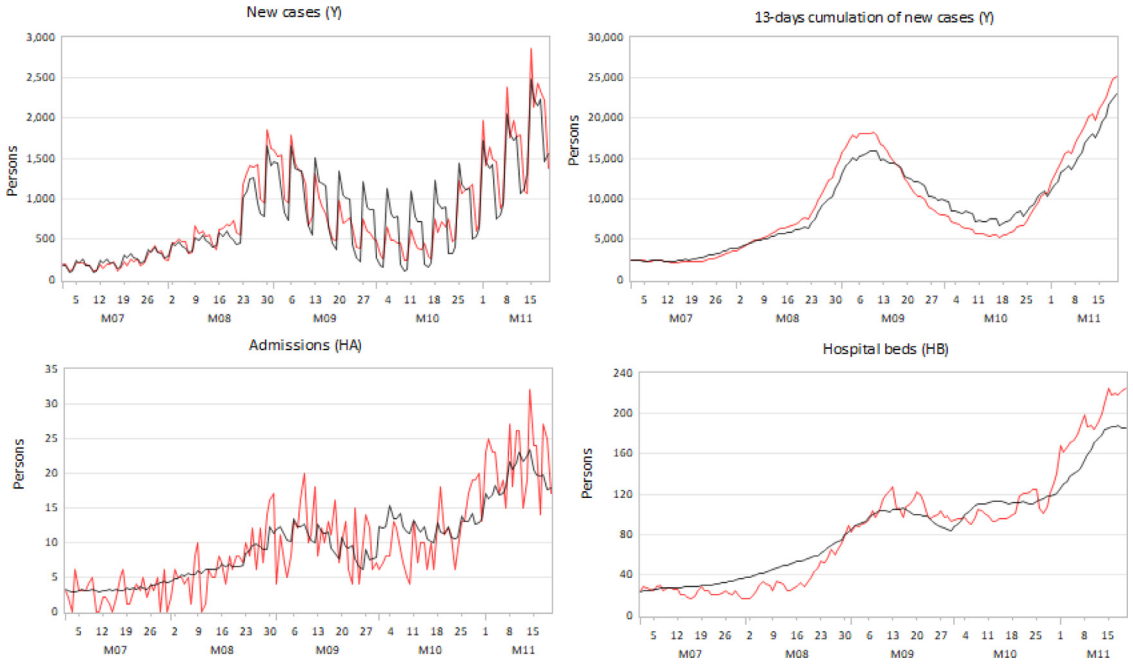


Fig. 2. Example of within sample dynamic simulation of CovidMod: estimated Eqs. (14), (15), (16) and two identities: (8) with $q = 13$ (in Y equation) and with $q = 15$ (in HA equation). Simulation start: 1 July 2021. End date of simulation: 20 November 2021. Red line graphs represent actual values and black lines simulated values. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

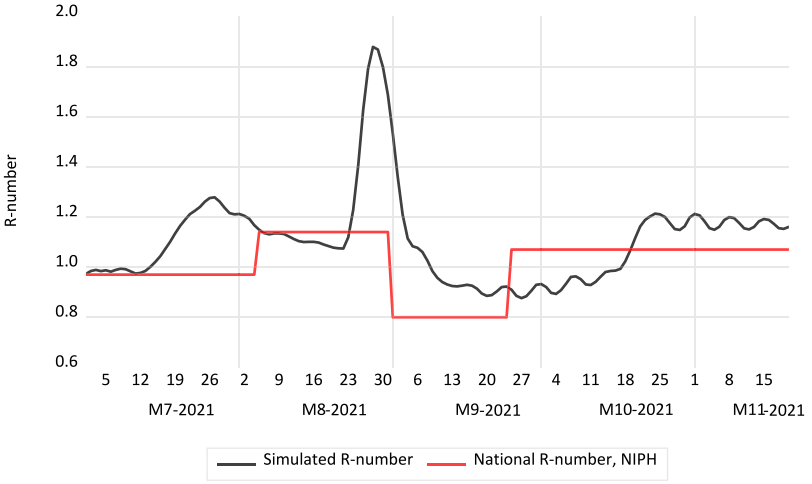


Fig. 3. Dynamic simulation of a time-varying $R_{t,4,4}$, using the simulated number of new cases in Fig. 2. Centred seven-day moving average. The national R-number for Norway produced by NIPH has been plotted as a reference.

number was reduced towards one and even a little below one. There was a new increase in October, to a level above one, and this level lasted for the rest of the simulation period.

The national *R*-number published by NIPH was visibly more constant over relatively long periods. It was estimated to be 0.97 until 4 August, and to have been 1.14 during the rest of that month. The lowest estimate at 0.8, was for the period from 1 September until 24 September. Starting 25 September, the national reproduction number was estimated to be 1.07.

It is not shown in Fig. 3, but the next change the NIPH estimated *R*-number came 13 December 2021. It was then estimated to be 0.79.[4]

[4] Source: Table 1 (page 21) in *Situational awareness and forecasting for Norway*, dated 5 January 2022. Between the start of the pandemic and the spring of 2023, 30 different R-numbers were calibrated and

4. Performance of the real-time CovidMod forecasts

The model simulation in Fig. 2 was based on actual values of the vaccination indicator, and also the empirically identified breaks due to changes in the virus, and changes in the NPI regimes decided by the government, based on the advice of the experts working in the National Institute of Public Health and in the The Norwegian Directorate of Health.

Fundamentally, when the framework was used in real time to forecast new cases, admissions to hospital and hospital beds, it was not possible to condition on future values of any exogenous explanatory variables. The missing modelling of effects of mutations of the virus that occurred after the origin date of a forecast, as well as effects of policy responses, therefore contributed to the forecast errors of the daily real time forecasts.

The development of the vaccine indicator VAC over the 21-days forecast horizon was also uncertain, and had to be forecasted. We did so by simple interpolation, drawing a line between the value of VAC at the date of origin of a forecast and the target level of the variable at a future endpoint. The target level and the endpoint were chosen in accordance with published plans for vaccination.

As mentioned, future effects of policy responses was not modelled in the version of the model that was used to produce daily forecasts. An approach to model the effects of policy responses is presented in a later section.

4.1. Comparison between CovidMod and NIPH forecasts

We first give a graphical illustration for a sub-set of forecasts, for new cases in the period from the beginning of November 2021 to the start of April 2022. In the first part of this calendar period, the NIPH was still publishing weekly forecasts of new cases with 21 days horizons obtained by the use of their in-house epidemiological model. That model is a stochastic SEIR-type model with a local transmission process in each municipality. The spread between municipalities is modelled by following how people travel between municipalities. The amount of travel between the different municipalities is based on mobile phone data. One of the key parameters in the model is the reproduction number, R, mentioned above.

In Fig. 4, actual numbers are shown in the red line graph. CovidMod forecasts are shown in the many black lines graphs. Each line graph represents a 21 days forecast, but since the forecasts were produced each workday, there is a considerable overlap. NIPH forecasts are shown in the green line graphs. The green line graphs are few in number. This is because publications of forecasts of COVID-19 incidence in the "Situational awareness and forecasting" reports became irregular in December, and in January 2022 they were discontinued. The last 21 days forecast of incidence was published in the report dated 5 January 2022.

The CovidMod forecasts appear to have been relatively good during November and December, although the scale of the graph may make the forecasts appear better than they were perceived in real time. Early in January 2022, the actual number of daily new cases was significantly higher than the forecasts with date of origin two or three weeks before. This is generally recognised as the time when the Omicron strain of the virus became dominant in Norway.

Fig. 4 shows that not only did the forecasts with date of origin late in December and in early January have a higher average than earlier forecasts, but also that there were higher variations from one forecast to the next. This feature is explained by greater variation in the initial conditions of the forecasts. The effects of Omicron and of the following non-pharmaceutical policy measures were also only gradually incorporated in the model, in particular in the light of the evidence of the forecast errors.

From 26 January 2022, persons who had had a booster dose, or who had received two doses of COVID-19 vaccine and had COVID-19 during the last three months, were no longer offered a confirmatory PCR test at a test centre after a positive self-test. It is credible that after this change in testing strategy, the data on new cases came to measure fewer new infections than before.

Nevertheless, the evolution of the number of new cases through February and March 2022 was quite well forecasted by CovidMod, at least judged by the plotted forecast graphs which are "on top of" the graph for actual number of new cases. Hence, the structural break in the data generating process of new cases that can be linked to the change in testing strategy, was not in itself reason to step away from attempting to model the data and to forecast the series.

Fig. 5 shows forecasts and actuals for hospital beds over the same period as in Fig. 4. 21 days forecasts from NIPH's National regional model were published throughout this period and are shown in the green line graphs in Fig. 5, while CovidMod forecasts are again shown in black line graphs. Note that the last observation of the time series for hospital beds (in red) is 22 March 2022 when the Norwegian Directorate of Health suddenly stopped the publication of those numbers.

The graph shows that both sets of forecasts have the "hedgehog" appearance which is familiar from economic forecasts. For example, the 14–21 day ahead CovidMod forecasts were systematically overshooting in November 2021, and undershooting in the development in mid-January and the first weeks of February. The same can be said about the NIPH forecasts. An additional remark that applies to those forecasts is that they sometimes got the level wrong "from the start"— the latest four forecasts for example. It can be an indication that evaluation of initial conditions may have been an important part of the preparation of these forecasts.

The sub-period of the pandemic covered by the previous graphs is interesting in itself, as it illustrates how the forecasts became adapted to a period of the pandemic characterised by a more infectious strain of the virus (Omicron) and a more open society, but also a gradually higher degree of vaccination of the population.

used in NIPH's forecasting model, cf. Table 1 in *Situational awareness and forecasting for Norway* 25 May 2023, Norwegian Institute of Public Health. In the period covered by our study, 24 R-numbers were used.

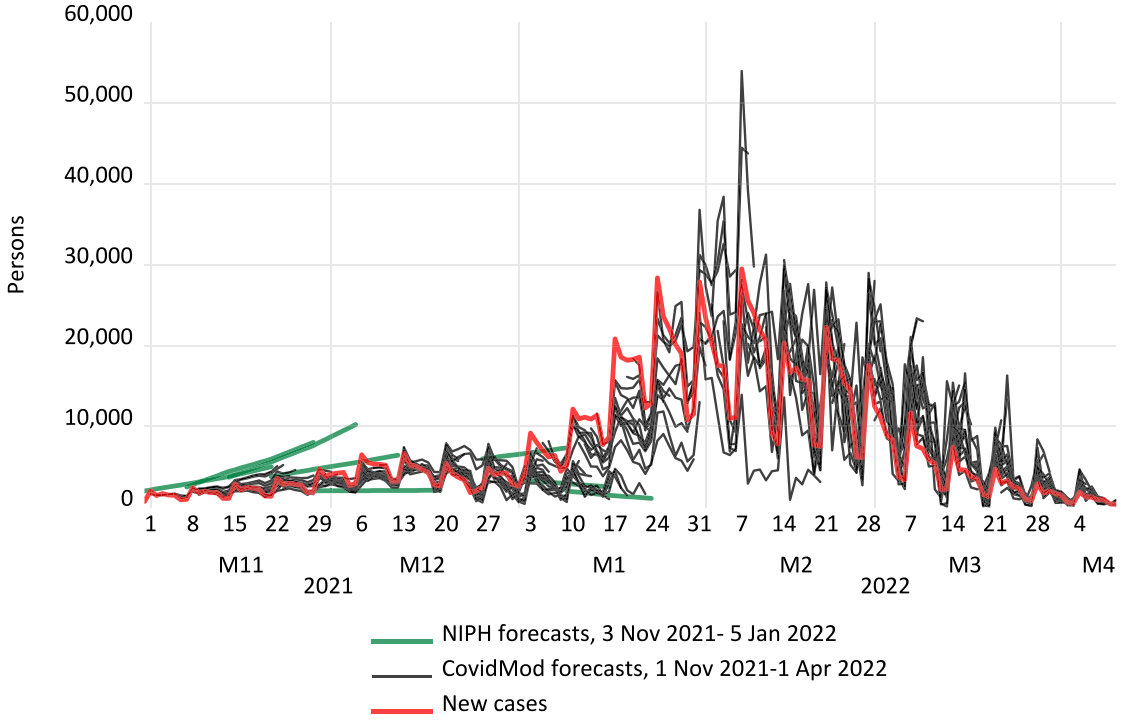


Fig. 4. Forecasted number of new cases (daily data) and forecasts from CovidMod and NIPH. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

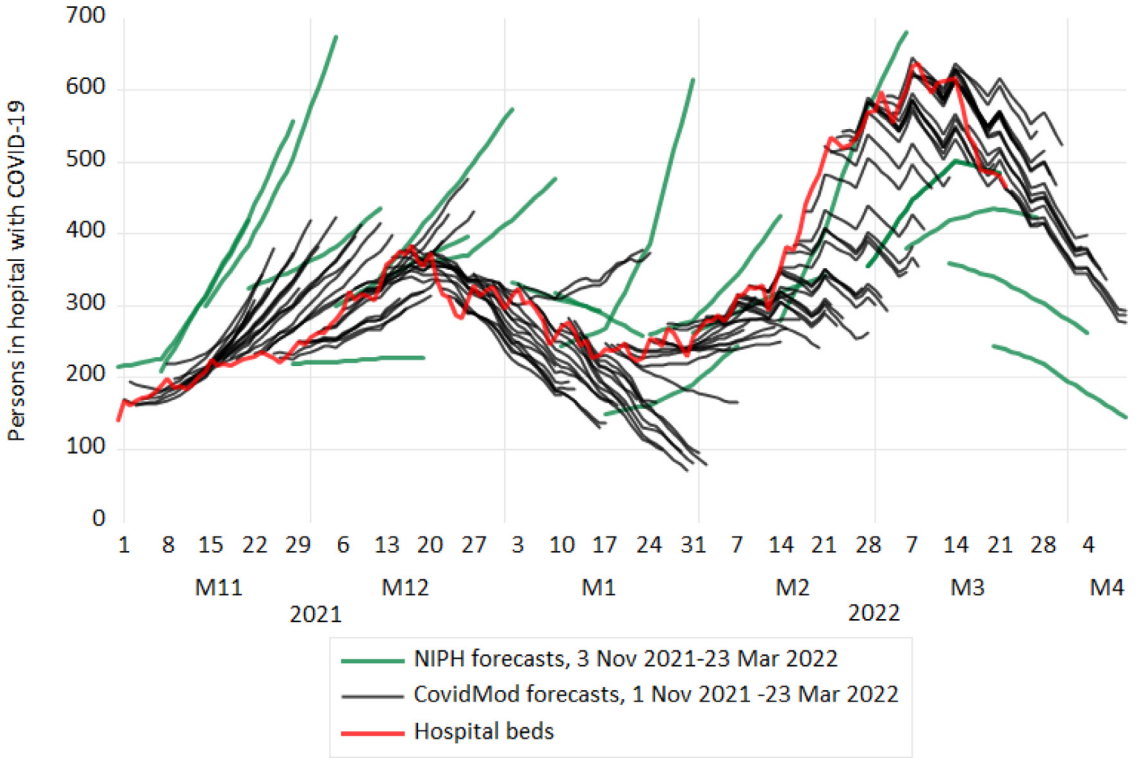


Fig. 5. Forecasted number of hospital beds and forecasts from CovidMod and NIPH. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

We next compare RMSFE (Root Mean Square Forecast Errors) from CovidMod and from the National and regional model of NIPH. The earliest forecasts have date of origin 19 March 2021 and the last originates from 1 December 2021. As mentioned above, this is when NIPH stopped the production of 21 days forecasts of incidence from their National and regional model.

For hospital beds the last forecasts are from 2 February 2022. Evaluation of later forecasts are not possible due to the stop in the publication of data for the number of persons in hospital with COVID-19. Hence, even though NIPH continued to publish forecasts of hospital beds, there are no forecast errors that can be evaluated.

All in all, this gave a sample of NIPH 33 forecasts errors for new cases and 44 for hospital beds. From the forecast record of CovidMod we have 178 forecast errors for new cases and 242 for hosital beds. As noted above, the CovidMod forecasts were prepared for each day in the working week, while NIPH forecasts were published weekly.[5]

Table 1 shows the RMSFEs as functions of forecast horizon. For new cases, the RMSFEs for the NIPH's model are (more than) two times larger than for CovidMod, for all horizons. For hospital beds, the CovidMod forecasts again have the lowest RMSFEs. However, except for the shortest horizons, the NIPH model's RMSFEs are not significantly higher numerically speaking.

The columns marked ARMA show RMSFEs of forecasts that have been constructed retrospectively. Therefore the ARMA forecasts were not affected by the "ragged edge" problem that we faced when using CovidMod to create the real time forecasts. Nevertheless the RMSFEs in the two ARMA columns are interesting for comparison, even though they may not be interpretable as feasible in real time forecasting. As the table shows, ARMA has the lowest RMSFEs for $h = 1$ and $h = 7$, but the difference from CovidMod becomes reduced for the two longer horizons. For New cases, $h = 14$, CovidMod is marginally better on the RMSFE measure.

Table 1
RMSFE for CovidMod forecasts with origin dates from 19 March 2021 to 1 December 2021 (new cases) and 2 February 2022 (hospital beds), and RMSFEs for NIPH forecasts with origins over the same period (n shows number of forecasts that the RMSFEs are based on).
Source: See appendix. The ARMA forecasts were constructed retrospectively using ARMA(8,4) for for New cases and ARMA(4,2) for Hospital beds.

Horizon	New cases			Hospital beds		
	CovidMod $n = 178$	NIPH $n = 33$	ARMA n=178	CovidMod n=242	NIPH n=44	ARMA n=242
$h = 1$	163	897	136	7	50	10
$h = 7$	429	1398	275	31	60	34
$h = 14$	630	1836	584	62	80	62
$h = 21$	852	2401	897	96	108	93

4.2. Comparison between CovidMod and Cardt forecasts

For new cases, we can make a RMSFE comparison with forecasts extracted from the multi-county COVID-19 forecasting project undertaken by Jennifer Castle, Jurgen Doornik, and David Hendry, at the University of Oxford.[6]

Those forecasts were for multiple countries and were produced by a automatised forecasting machinery developed by the research team, see Castle, Doornik., and Hendry (2021a), Doornik et al. (2020). The forecasted variables were confirmed cases and deaths. The statistical forecasting device, dubbed Cardt, is highly adaptive and it rapidly adjusts to the latest information in the data, handling both stochastic trends and abrupt shifts, (Castle, Doornik, & Hendry, 2021b). It is based on decomposing the data into a trend, seasonal and irregular component, and then forecasts the components separately before aggregating.[7]

The research group in Oxford produced real time forecasts from 20 March 2020 to 16 June 2022. This gives a longer period of overlap with our forecasts than was the case for the forecasts made by NIPH. The Cardt forecasts were for the total number of confirmed cases. However, because of the definition (7), we can obtain new cases from the Cardt forecasts as:

$$Y_t^{Cardt} = S_{Y,t}^{Cardt} - S_{Y,t-1}^{Cardt} \tag{18}$$

where $S_{Y,t}^{Cardt}$ were the forecasts published on the internet. The difference between these forecasts and the NIPH data for new cases by date of registration gives the Cardt forcast errors for Norway. As noted above, CovidMod used data by date of specimen collection. That difference is however not large enough to undermine the comparability of the two sets of forecast errors.

The multiple-country Cardt forecasts were short-run, with forecast horizons from one to seven days. Fig. 6 shows graphs of new cases together with 7-days forecasts, for the same calendar period as in Fig. 4. There is little visible difference between the CovidMod and Cardt forecasts in November 2021 and the first part of December 2021. As was shown immediately above, the CovidMod forecast became more volatile in the winter of 2021, and the same can be said about the Cardt forecast graphs in Fig. 6.

As shown in Table 2, the number of forecasts produced over the whole overlapping period was 265 CovidMod forecasts, and 83 Cardt forecasts.

Compared to the new cases part of Table 1, the RMSFE numbers in Table 2 are much higher. This is because in Table 1, comparability with NIPH's forecasts of new cases meant that the forecast errors came from a period when new cases were less numerous than later in 2021 and during the winter of 2022. In the comparison with Cardt in Table 2 the errors from that period could be included.

In Table 2 there are fewer cases than in Table 1 where the ex-post ARMA forecasts have lower RMSFEs than the two real time forecasts. For $h = 14$ and $h = 21$ CovidMod seems to have performed better. Since one difference between the two tables is that Table 2 contains several

[5] There were a few examples of two weeks between forecasts, in connection with holidays.

[6] https://www.doornik.com/COVID-19/

[7] Cardt is short for Calibrated Average of Rho, Delta and THIMA.

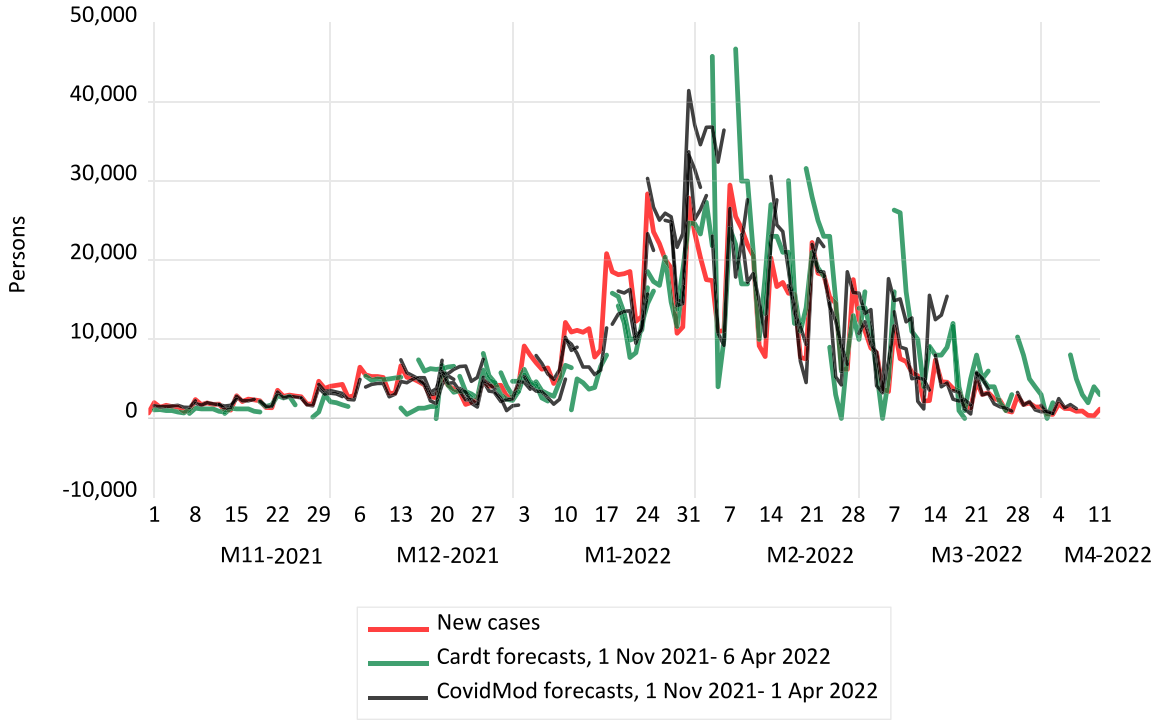


Fig. 6. Forecasted number of new cases and forecasts from Cardt and CovidMod (7-days horizon).

Table 2
RMSFE for new cases, CovidMod and Norway forecasts from Castle, Doornik and Hendry multiple-countries and regions forecast, Cardt. Forecast origins from 19 March 2021 to 1 April 2022 (CovidMod, and ARMA), and from 18 March 2021 to 6 April 2022 (Cardt).
Source: See appendix. The ARMA forecasts were constructed retrospectively, using ARMA(8,4).

Horizon	CovidMod $n = 264$	Cardt $n = 83$	ARMA $n = 264$
$h = 1$	1658	5928	1051
$h = 2$	1528	3485	1317
$h = 3$	1385	2845	1500
$h = 4$	1784	1890	1922
$h = 5$	1829	2174	2252
$h = 6$	2350	2662	2385
$h = 7$	2661	2666	2590
$h = 14$	4181	–	4599
$h = 21$	6852	–	8131

more forecast errors from the Omicron-period of the pandemic, the interpretation is that there were many large ARMA forecast errors in that period, also compared to CovidMod's real time forecasts

Table 2 shows that there are some differences between the RMSFEs of the two forecasting devices for the short horizons ($h = 1, 2$ and 3). However, there is in practice no difference between the RMSFEs for the seven days forecasts.

5. Extending the model with endogenous effects of policy interventions

As noted above, future effects of policy responses were not modelled in the operative version of the model that was used to produce the daily forecasts. Instead, the practical method was to adjust the model ex post, by estimating the effects of a policy change soon after it was included.

In a pandemic, interventions can take many forms. However, from the start of the COVID-19 pandemic in the winter of 2020, and until the spring of 2022, policy interventions were consistently rationalised by the number of hospital beds occupied by patients with COVID-19 threatening to overwhelm the national health service system. This, it appears, represents a regularity in the data generating process that could made subject to econometric treatment.

Here, we present one way to include in the model the effects of policy responses. As noted, interventions can take many forms, and we only try to capture the effects of the policy actions that can be modelled by assuming that the policy effects variable is the number of hospital beds, and the deviation of that variable from a target level.

Fig. 7 illustrates some of the differences between CovidMod, as illustrated by Fig. 1, and the model with endogenous policy response effects, which is a separate node in Fig. 7. Policy response effects should be interpreted broadly, as representing all policies that can be modelled as responses to the evolution of hospital beds with COVID-19 patients, both vaccinations and non-pharmaceutical interventions.

In the flowchart figure representing CovidMod-STR, (exogenous) changes in the virus is a separate node. Mutations can influence the pandemic's progress in many ways (as indicated by the dotted lines), but in our model, not through direct affectation of policy response effects. We

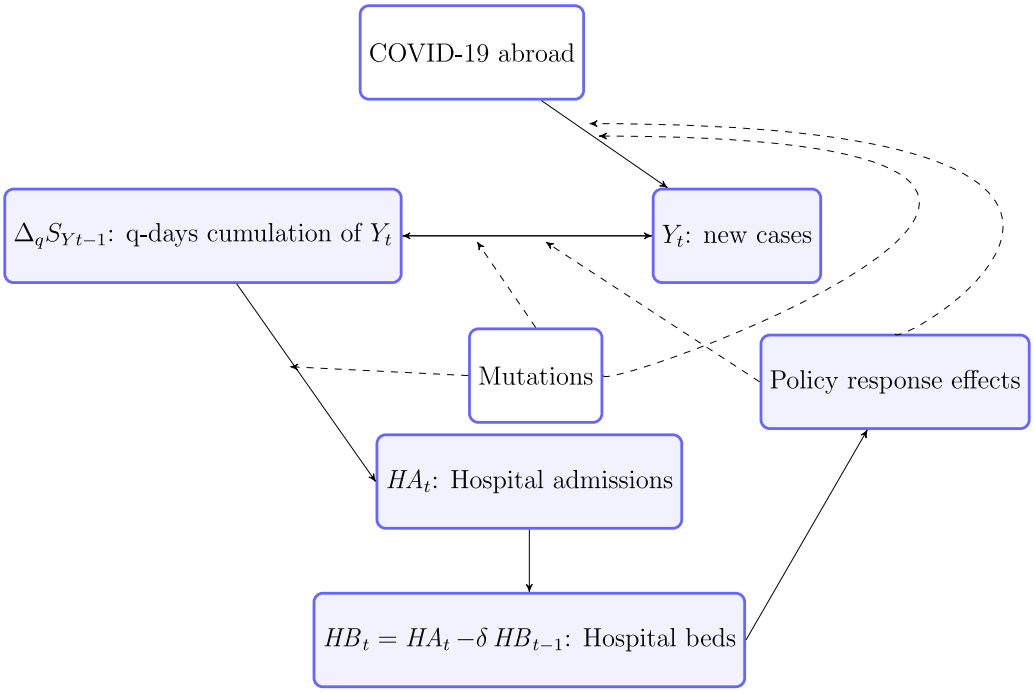


Fig. 7. Variables and relationships in CovidMod-STR.

picture that relationship as more indirect, through how new strains of the virus may affect hospital admissions and hospital beds (relative to target).

We now turn to some of the details about the implementation of the idea about endogenous policy response in the model, before we illustrate the forecasting potential of a fully specified and estimated version of the model.

5.1. Implementing endogenous policy responses

In modelling the effects of endogenous policy responses, we divide the breaks or the switching part of the model, into exogenous regime changes D_t, like mutations, and effects of endogenous policy regime changes G_t, like lockdowns. The endogenous changes are seen as responses to perceived threats to the health care system, represented in the model by the number of hospital beds relative to a target level.

In order to illustrate the approach, consider the following stylized version of Eq. (11), where $X_t = (S_{Y,t-1} - S_{Y,t-p})$:

$$Y_t = \beta_0 + \beta_1 X_t + \beta_2 X_t D_t + \beta_3 X_t G_t + u_t \tag{19}$$

An exogenous event, or shock, makes the binary variable D_t change the effect of X_t on Y_t from β_1 to $\beta_1 + \beta_2$.

To model the endogenous policy effects G_t, we use the standard Smooth Transition Regression model (STR), see e.g., van Dijk et al. (2002) and Teräsvirta et al. (2010, Ch. 3.4) for overviews. As is common, the transition function is the logistic specification

$$G_t\left(HB_{t-k};\theta, HB^*\right) = \frac{1}{1+\exp\left[-\theta\left(HB_{t-k} - HB^*\right)\right]}. \tag{20}$$

In our application, the transition function has a policy target variable as argument. The policy variable is taken to be hospital beds HB_{t-k}, with a threshold value of HB^*.

Eq. (20) then implies: If $\left(HB_{t-k} - HB^*\right) \to -\infty$

$G_t \to 0$;

if $HB_{t-k} = HB^*$, then

$G_t = 0.5$;

and with $\left(HB_{t-k} - HB^*\right) \to \infty$

$G_t \to 1$.

With a very high smoothness parameter θ, the effects of switching between zero or full policy effects can approximate a binary variable—see van Dijk et al. (2002, sec. 2).

The parameter θ and the threshold value HB^* of the transition function are estimated by non-linear least squares, with starting values based on a grid search, conditional upon the linear parameter estimates — see van Dijk et al. (2002, sec. 5.2) and Teräsvirta et al. (2010, Ch. 16.3.3)

To continue the illustration: Imagine a mutation, represented by $\beta_2 D_t > 0$ in (19), leading to large increases in Y and HB. The policy response might be a lockdown or an increase in vaccinations, bringing the infections down, with part of the effect captured by $\beta_3 G_t < 0$. For illustration purposes, assume that $D_t = G_t = 1$. If then $\beta_3 = -\beta_2$, the effects of policy interventions mitigate the effects of the virus mutation.

To test out the suggested approach with endogenous policy effects, we estimated the model with data until 5

January, 2022, and evaluated the forecasting performance over the period 7 January, 2022–29 January, 2022.

5.1.1. The STR-model equation for new cases Y_t

The equation for Y_t is estimated as:

$$\begin{aligned}\hat{Y}_t &= \underset{(0.004)}{0.060}\left(S_{Y,t-1}-S_{Y,t-14}\right)\\ &+\underset{(0.002)}{0.007}\left(S_{Y,t-1}-S_{Y,t-14}\right)D_t\\ &-\underset{(0.004)}{0.007}\left(S_{Y,t-1}-S_{Y,t-14}\right)G_t\\ &+\sum_{i=0}^{8}\hat{\alpha}_i^{\ddagger}\Delta Y_{t-1-i}+\text{ residual}\\ T &= 15.2.2020-5.1.2022,\ 691\text{ obs.}\end{aligned} \tag{21}$$

where

$$D_t = f(D_{It}+D_{\alpha t}+D_{\delta t}+D_{ot})$$

$$G_t = \frac{1}{1+\exp\left[-\underset{(0.032)}{0.029}\left(HB_{t-5}-\underset{(48.575)}{295}\right)\right]}.$$

HAC standard errors are reported in parentheses below the estimates.

The step dummies in $D_t = f(D_{It}+D_{\alpha t}+D_{\delta t}+D_{ot})$ represent exogenous breaks due to the occurrence of imported infections and the Alpha, Delta, and Omicron mutations, respectively:

$$\begin{aligned}D_{It} &= 1 \text{ from } 09.03.2020\\ D_{\alpha t} &= 1 \text{ from } 02.03.2021\\ D_{\delta t} &= 1 \text{ from } 16.07.2021\\ D_{ot} &= 1 \text{ from } 28.12.2021\end{aligned}$$

As in the model used in the real time forecasts, new cases are modelled as a function of the infection level in the population, which is however not observable. In Eq. (21), and in line with the formulation in the CovidMod, the change in the accumulated level of cases over a two-week period, $(S_{Yt-1}-S_{Yt-14})$ is used as a an indicator of the infection level. For the policy effects function G_t, the threshold value $HB^{\star}$ is estimated to be 295 hospital beds. Note that the effects of exogenous pandemic shocks D_t are mitigated by the endogenous policy effects G_t. The chosen specification, with H_{t-5} in the argument of the function, may seem as a short time lag for a lockdown to start having effects on new cases. However, it was the lag-length that gave the best fit on the estimation sample.

Although not significant, the steepness parameter of the transition function θ is estimated to be close to 0.03, indicating that the policy effects were small as long as the policy target variable was below the treshold.

Fig. 8 shows the number av patients in hospital with COVID-19, HB_t, on the left scale, and the values of the smooth transition function G_t, on the right scale.

As shown in Fig. 8, the policy effects never completely counteract the exogenous shocks, since the G_t function never reaches the value 1.

To complete the model, the estimated versions of *HA* (hospital admissions) and *HB* (hospital beds) equations follow.

$$\begin{aligned}\widehat{HA}_t &= \underset{(0.0001)}{-0.0006}\left(S_{Y,t-3}-S_{Y,t-9}\right)D_{o(t-6)}\\ &+\underset{(0.001)}{0.012}Y_t\\ &-\underset{(0.001)}{0.007}Y_t\times D_{1,t}\\ &-\underset{(0.001)}{0.002}Y_t\times D_{2,t}\\ &+\sum_{i=j}^{3}\alpha_j^{\ddagger\ddagger}HA_{t-j}+\text{ residual}\end{aligned} \tag{22}$$

$$\begin{aligned}\widehat{HB}_t &= HA_t+\underset{(0.004)}{0.901}HB_{t-1}\\ T &= 14.7.2020-5.1.2022,\ 541\text{ obs.}\end{aligned} \tag{23}$$

$D_{1,t}$ = Step-dummy, 1 from 28 April 2021.

$D_{2,t}$ = Step-dummy, 1 from 15 November 2021.

Although qualitatively similar in terms of variables, the specification in (22) has a more complex structure of the effects of new cases, Y_t, than (15). A possible interpretation of the dummies might be that more young people became infected, but with a lower frequency of hospital admittance.

Eq. (23) is almost the same as (16).

5.2. Forecasting with the non-linear model, CovidMod-STR

In order to investigate the forecasting potential of the model with effects of endogenous policy interventions, we simulated the model dynamically with 7 January 2022 as the first forecast period and a (good) three weeks ahead (29 January, 2022 was the last solution period). The simulation did not condition on any exogenous variables in the simulation period, hence mimicking a real-time forecasting situation.

Fig. 9 shows the three-week dynamic forecasts of the STR-version of CovidMod, with 7 January 2022 as the first day forecasted:

As illustrated by the graphs in Fig. 4, the pandemic had "changed gear" in the weeks before Christmas 2021, and the CovidMod forecast then struggled to adapt to the increased numbers of new cases and admissions to hospital. Fig. 9 indicates that CovidMod-STR would have performed significantly better. Not only for new cases, but also for new hospital admissions *HA* and number of hospital beds *HB*. $\left(S_{Y,t-1}-S_{Y,t-14}\right)$ is also shown (in the second panel). Hence, taken together, the estimation results for the STR-model for new cases, and the dynamic simulation, show that forecasting with a model that includes the effects of certain policies to contain the virus, is a feasible method.

Figs. 10 and 11 plot the CovidMod-STR (mean) forecasts for new cases and for hospital beds, together with the NIPH and CovidMod forecasts from the first week of January 2022. The simulation results of the model with endogenous policy effects appear to be unbiased for new cases, for most of the period, while both CovidMod and NIPH under-forecasted the actual number of new cases in January.

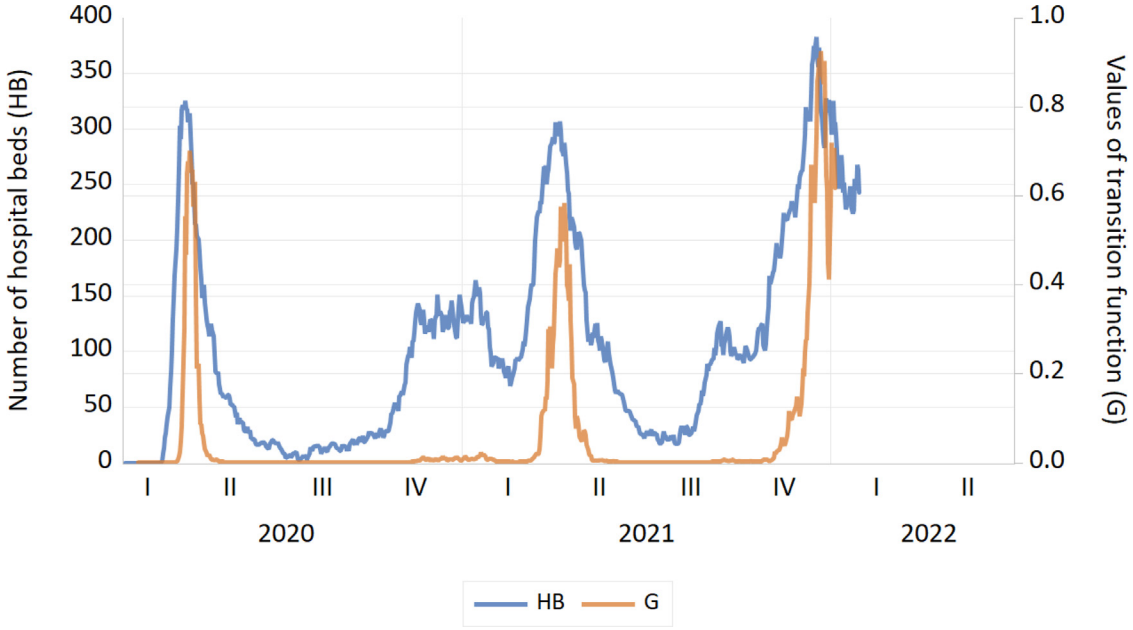


Fig. 8. Hospital beds HB_t (left scale) and smooth transition function G_t (right scale).

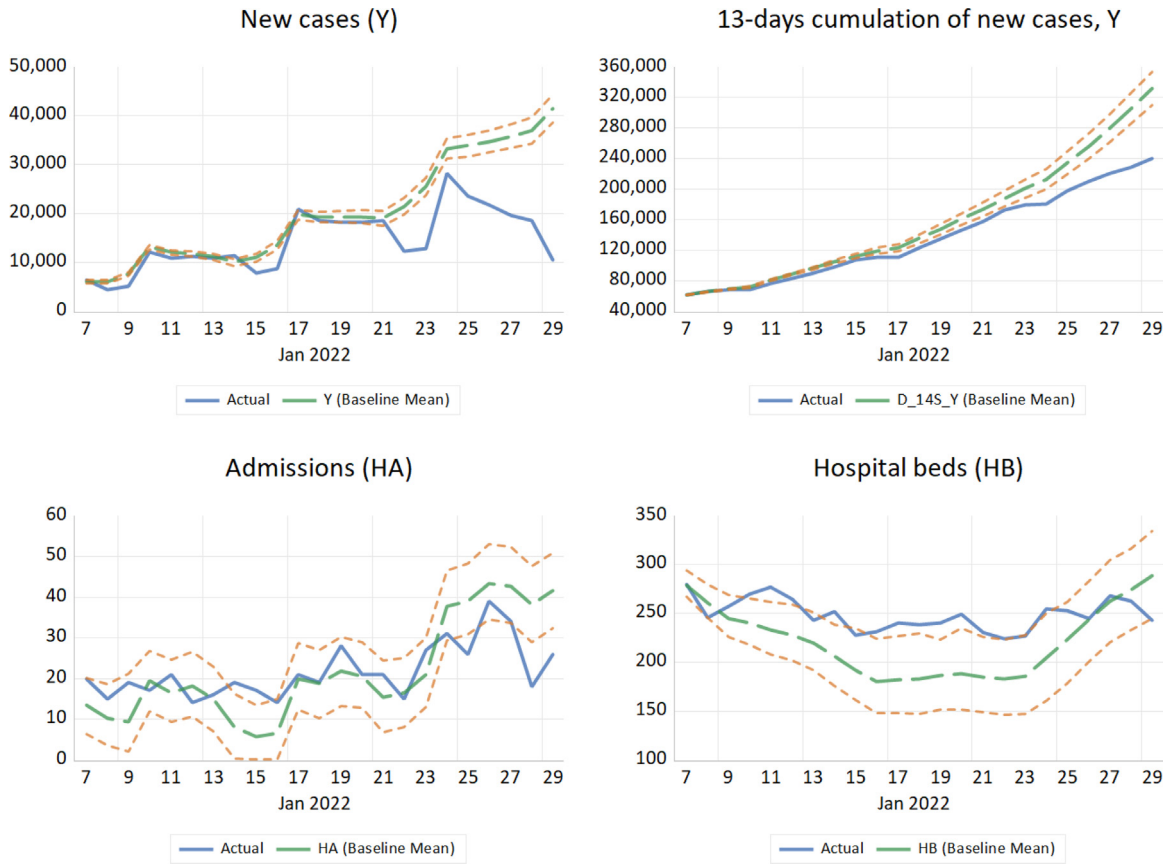


Fig. 9. The CoviMod-STR forecasts and realizations. Line graphs represent actual values, dashed lines represent simulated values and dotted lines represent upper and lower bounds of 90% prediction intervals. The intervals are based on 1000 replications of bootstrapped innovations with a non-diagonal covariance matrix. The intervals are simulated without parameter uncertainty.

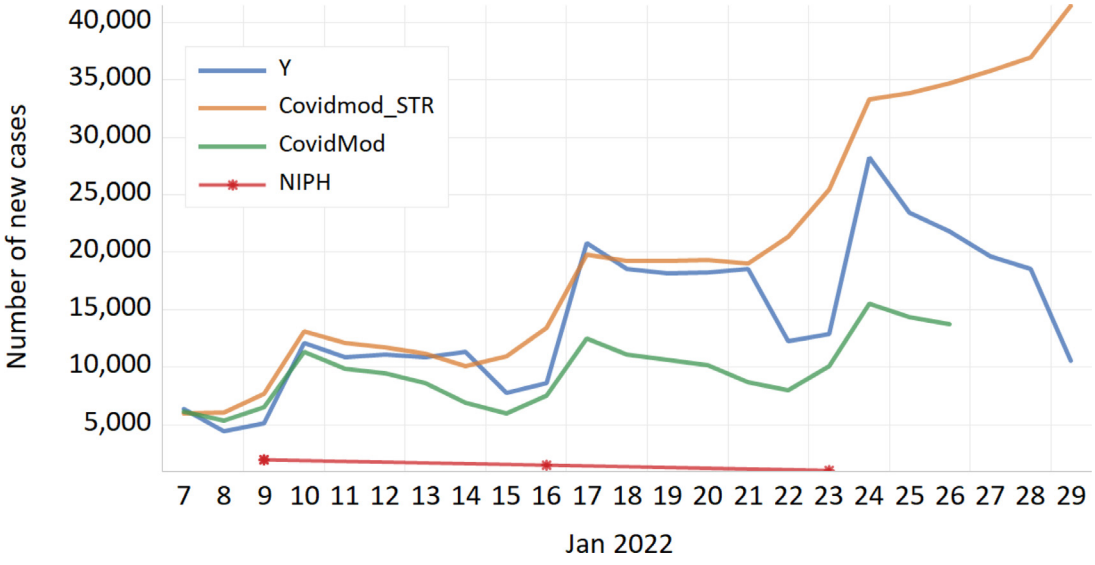


Fig. 10. New cases. NIPH COVID-19 modelling team Week 1, 5 January 2022. Table 2. Linear interpolation between 7, 14 and 21 days ahead forecasts from forecast origin date.
Source: "Situational awareness and forecasting for Norway".

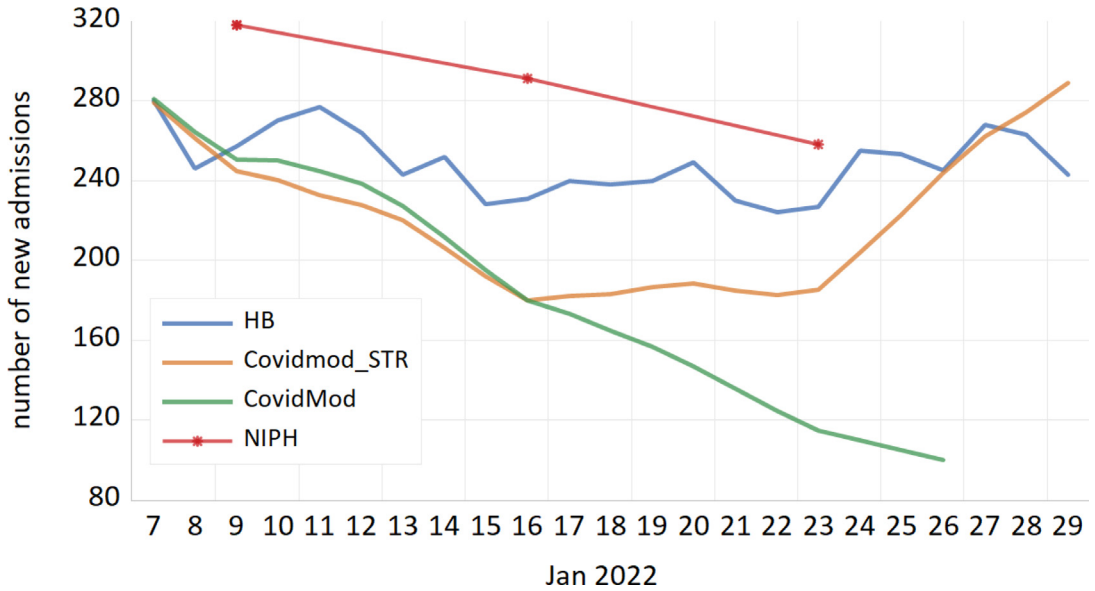


Fig. 11. Hospitalisations. NIPH COVID-19 modelling team Week 1, 5 January 2022. Table 2 and Fig. 5. Linear interpolation between 7, 14 and 21 days ahead forecasts from forecast origin date.
Source: "Situational awareness and forecasting for Norway".

When we look at the results for hospital beds, Fig. 11, the picture is more complex. The two sets of CovidMod forecasts have biases (albeit insignificant when the uncertainty is taken into account), but we note that unlike the CovidMod forecasts, the forecasts of the STR version of the model recoup the level of the actual number of hospital beds at the end of the forecast period.

Table 3 shows RMSFEs for new cases and for hospital beds, for the non-linear version model (CovidMod-STR) and the other forecasting methods considered. The RMSFEs in Table 3 are for the one-off forecasts plotted in Figs. 10 and 11, i.e., similar to having a sample of forecasts equal to one ($n = 1$) in Table 1. The table shows that for new cases, it is the STR model that has the lowest RMSFEs for $h = 7$ and $h = 14$. However, for the three-week horizon, the RMSFE of CovidMod is a little lower than the RMSFE of the STR version of the model. As can be gleaned from the graph of the NIPH forecast in Fig. 10, which was

Table 3
RMSFEs for different forecasts with date of origin in the first week of January 2022.
Sources: See Tables 1 and 2. NIPH, Situational awareness and forecasting for Norway:

Horizon	New cases				Hospital beds		
	CovidMod-STR	CovidMod	NIPH	Cardt	CovidMod-STR	CovidMod	NIPH
$h = 7$	1305	1327	7572	4047	27.0	19.8	53.2
$h = 14$	1912	4484	83221		41.4	50.2	52.9
$h = 21$	7700	7378	12557		48.6	106.7	46.3

practically out of scale with the actual numbers of new cases in this period, the corresponding RMSFEs were very large compared with those of CovidMod-STR in particular.

For completeness, the New cases part of the table also includes the RMSFE for the seven day Cardt-forecast with forecast origin in the first week of January 2022.

In the Hospital beds part of the table we see a somewhat different picture. CovidMod got the lowest RMSFE for the $h = 7$ horizon, while for $h = 14$ the RMSFE of CovidMod-STR was the lowest of the three methods. Interestingly, for $h = 21$ the RMSFE of the NIPH forecast was the lowest. This is as one could expect from the graphs in Fig. 11, which show that although the NIPH forecast in question was way off the mark for the short-horizons, it nevertheless ended up quite close to the actual number of hospital beds three weeks after the production of the forecast. However, the RMSFE of CovidMod-STR was only marginally larger for the $h = 21$ horizon.

5.3. Extending the estimation sample of the STR-model

It is of interest to check how robust the estimation result for the STR-equations are with respect to extensions of the sample. In this section, we use the longest estimation sample that still allows comparison with actual values of both new cases and hospital beds, namely a sample that has 1 March as the last observation.

The results for the Y_t equation are:

$$\begin{aligned}\hat{Y}_t &= \underset{(0.004)}{0.081}\left(S_{Y,t-1} - S_{Y,t-14}\right) \\ &\quad - \underset{(0.0013)}{0.0002}\left(S_{Y,t-1} - S_{Y,t-14}\right) D_t \\ &\quad - \underset{(0.0009)}{0.008}\left(S_{Y,t-1} - S_{Y,t-14}\right) G_t \\ &\quad + \sum_{i=0}^{8} \hat{\alpha}_i^{\ddagger} \Delta Y_{t-1-i} + \text{ residual} \\ T &= 15.2.2020 - 1.3.2022, \ 746 \text{ obs.}\end{aligned} \tag{24}$$

where

$D_t = f(D_{It} + D_{\alpha t} + D_{\delta t} + D_{ot})$ is defined above, and:

$$G_t = \frac{1}{1 + \exp\left[-\underset{(10932005)}{15.002}\left(HB_{t-5} - \underset{(56886.11)}{311}\right)\right]}.$$

HAC standard errors are reported in parentheses below the estimates. Compared to (21) the effect of imported infection and virus mutants, D_t, is no longer significant. The policy effect function G_t is still just as important, but is now resembling an indicator variable, indicated by a slope θ estimate of 15. The steepness could be an explanation for the large standard errors—see Teräsvirta et al. (2010, Ch. 16.3.3) for a thorough explanation.

The re-estimated versions of *HA* (hospital admissions) and *HB* (hospital beds) equations are as follows:

$$\begin{aligned}\widehat{HA}_t &= \underset{(0.0000316)}{-0.0000196}\left(S_{Y,t-3} - S_{Y,t-9}\right) D_{o(t-6)} \\ &\quad + \underset{(0.001)}{0.004} Y_t \\ &\quad - \underset{(0.001)}{0.002} Y_t \times D_{1,t} \\ &\quad - \underset{(0.0005)}{0.0014} Y_t \times D_{2,t} \\ &\quad + \sum_{i=j}^{3} \alpha_j^{\ddagger\ddagger} HA_{t-j} + \text{ residual}\end{aligned} \tag{25}$$

$$\begin{aligned}\widehat{HB}_t &= HA_t + \underset{(0.007)}{0.895} HB_{t-1} \\ T &= 14.7.2020 - 1.3.2022, \ 596 \text{ obs.}\end{aligned} \tag{26}$$

Again, HAC standard errors are reported in parentheses below the estimates. $D_{1,t}$ and $D_{2,t}$ are the step-dummies defined above, in connection with (22). In the HA_t equation, the estimated coefficient of Y_t has been reduced compared to (22), which is not surprising given what was noted in connection with CovidMod about the influence of data from 2020 for the estimation results for this coefficient.

Fig. 12 shows the 21 days dynamic forecasts of this re-estimated version of CovidMod-STR, with 2 March 2022 as the first day forecasted:

Interestingly, the model still seems to be relevant over the extended sample, where 21 days have been kept for forecasting, as before. The forecasts of new cases now appear more volatile, approaching the zero lower bound in some cases.

6. Discussion

From model-based forecasting in economics, we are used to the fact that structural breaks in the economy is the main reason why forecasts become inaccurate and why downright forecast failures are not unusual, see Clements and Hendry (2002), Eitrheim et al. (2002), Falch and Nymoen (2011) and Bårdsen et al. (2017) amongst others.

COVID-19 time series are typical examples of data generating processes with many breaks, as a result of changes in the virus, lockdowns, voluntary social distancing and vaccinations. It is likely that forecasters who do not adjust models and model based forecasts in response to evidence

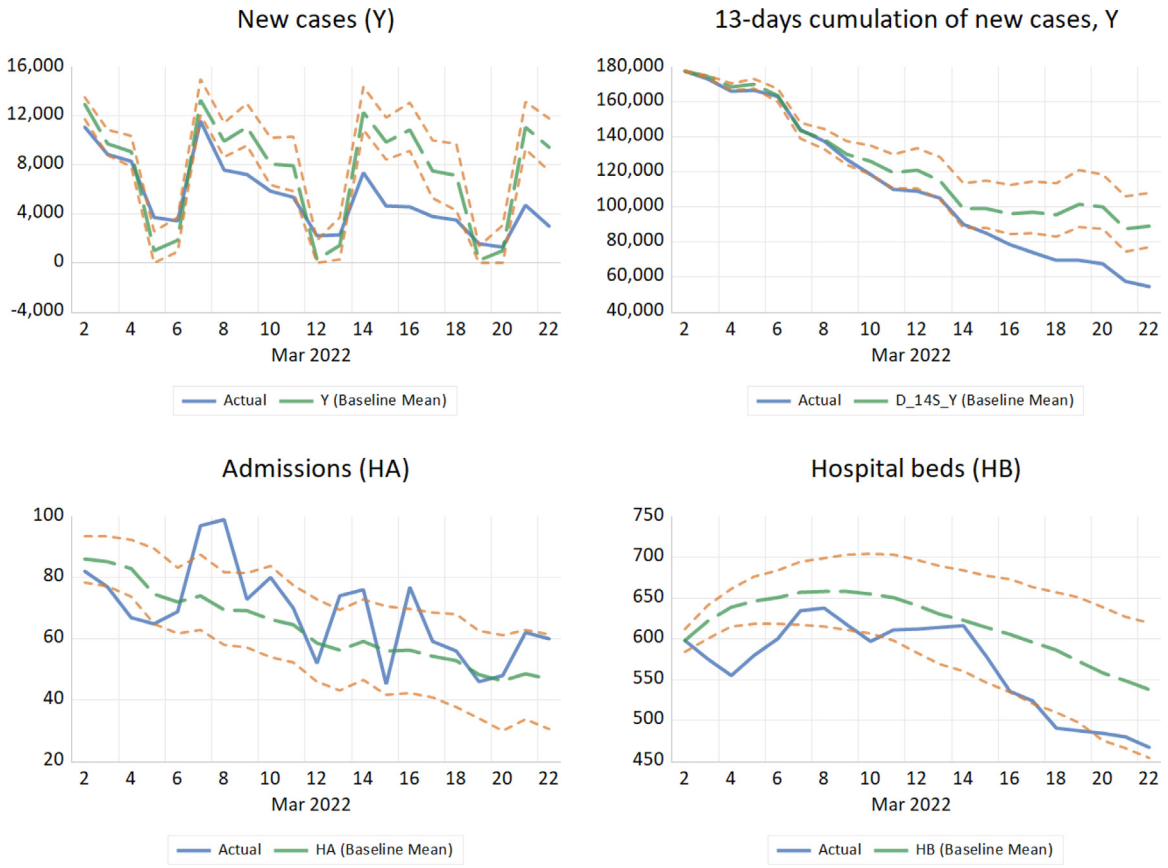


Fig. 12. The CoviMod-STR forecasts and realizations after estimation on the extended sample. Line graphs represent actual values, dashed lines simulated values and dotted and lines represent upper and lower bounds of 90% prediction intervals. The intervals are simulated without parameter uncertainty.

about such changes, will produce forecasts that are less accurate than they need to be.

Exogenous breaks that occur after the date of origin of a forecast are near impossible to predict. In practice, the best one can do is to include the breaks into the information set to be used for later forecasts, as soon as they have become detectable in the recorded forecast errors. This type of forecast adjustment, using observed forecast errors near to the date of the forecast origin to form intercept corrections, can improve forecast accuracy on bias measures in particular, although there may be some cost in forecast-error variance, see Clements and Hendry (1999, Chapter 6). Hence, it is good practice to study the observed forecast errors with the purpose of using past errors as data that may robustify the forecasts, Nymoen (2019, Chapter 12).

In this paper we have presented a forecast framework and model, dubbed CovidMod, which was used for real time forecasting of new cases, admissions to hospital and of hospital beds, and which allowed evidence of breaks to change key parameters of the model and to form a basis for intercept correction. In comparison, the SEIR-type model which was used to produce the 21 days ahead forecasts published by NIPH may have been less adaptive to breaks in the data generating process of the forecasted variables, even though the model was re-calibrated during the pandemic. This then, we suggest, is a main explanation why, based on RMSFE comparisons for forecast horizon between one and 21 days, the forecasts of CovidMod performed better than the NIPH forecasts, over the available comparison period.

Given that a main problem that afflict both economic and pandemic forecasting are the changes in the data generating processes that we "don't know that we don't know", and in particular shifts in the deterministic terms in the relationships between variables, statistical forecasting methods that adapt quickly can be expected to perform relatively well when compared with forecasts from structural models that rely on constancy of parameters in the forecast period. The Cardt forecasting device is a particular special case of a rapidly adapting method, that has been shown to work well in forecast competitions, (Castle, Doornik., & Hendry, 2021a). As noted above, Cardt has been used efficiently to produce real-time forecasts of COVID-19 confirmed cases and deaths for multiple countries, (Doornik et al., 2022).

In this paper we compared Cardt forecasts for Norway (new cases) with the real-time CovidMod forecasts. As the Cardt forecasts aimed at providing policy makers with projections of what might happen in the week ahead, the

RMSFE comparison was for a seven day horizon. For the available comparison period there was very little difference between the one week ahead RMSFEs of CovidMod and Cardt. In turn, this meant that also Cardt worked better than the forecasts produced by the SEIR-type model used by the National institute of public health. Since we have focused on the comparison of forecasts from different model types, it is perhaps worth stating that the overall outcome does not entail that the most adaptive or "robustified" model should be used for policy. Epidemiological models do not lose their scientific basis if they become inferior in a forecast comparison. They remain invaluable for rational assessment and understanding of what is going on during a pandemic. Development of operational a SEIR-model before, during and after COVID-19 was therefore much needed, and was also the route that the Norwegian modellers took.[8]

However, during the pandemic the accuracy of short-term forecasts became a concern in its own right. Forecasts were in demand as an aid in the decisions that were made about lockdown policy and in the allocation of scarce public health resources. In this situation, it is possible that the use of more than one model type for forecasting could have improved the quality of the foretold near future.

A more general question might be raised about the usefulness of short horizon forecasts in policy making. As just noted, during the pandemic many policy decisions were rapidly decided and implemented, meaning that also seven day ahead forecasts were regarded as practically useful. In Norway, the regular frequency of the publication of the NIPH forecasts was weekly. As noted earlier, the horizon of the NIPH forecasts was 21 days, and the institute stated explicitly that "We believe that it is not possible to make good long-term forecasts for the COVID-19 pandemic. Therefore, our forecasts are limited to 3 weeks ahead".[9] In an early offering, with real-time forecasts, Petropoulos and Makridakis (2020) argued for a 10 day forecast horizon.

Another contribution that we make in the paper is to model effects of policy induced endogenous breaks as a function of a policy target variable by the use of a smooth transition function integrated in the system of equations. The example of forecasting with the nonlinear model CovidMod-STR suggests that this approach seems to work in an operational context and might therefore be a useful tool in forecasting where policy reaction effects are relevant. This is the case for pandemics. Li et al. (2023) provides an impressive example of a forecasting model with policy effects within an epidemologic model framework. More generally in the social sciences, it is realistic that in the forecast period, policy decisions can be made that affect the outcome of the forecasted variable. Sometimes a policy maker may also act on the forecast itself, rather than on the outcome of the forecasted variable. This idea has attracted the interest of both theorists and practitioners, see the early contributions of Johansen (1972) and Granger (1973). Granger characteristically foresaw the difficulties of forecasting, and advised that forecasters and policy makers, or at least their advisers, should recognize the advantages of working in conjunction rather than in isolation.

[8] https://www.fhi.no/en/id/corona/coronavirus/coronavirus-modelling-at-the-niph-fhi/

[9] See https://www.fhi.no/en/id/corona/coronavirus/coronavirus-modelling-at-the-niph-fhi/, where it is also stated that the institute is also responsible for intermittent scenario analyses (not forecasts) with longer than three weeks time horizon.

CRediT authorship contribution statement

Gunnar Bårdsen: Conceptualization, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Ragnar Nymoen:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We are pleased to acknowledge funding from Norwegian Research Council project number 324472, on 'Model invariance and constancy in the face of large shocks to the Norwegian macroeconomic system'. We thank three referees and an associate editor for comments, both critical and constructive. Earlier versions of this work have been presented at the 25. Dynamic Econometrics Conference, London 13-14 April 2023; at the "Network Meeting for Modellers" arranged by Norwegian Institute of Public Health, 3 November 2022; and at the start up workshop of the project "Model invariance and constancy in the face of large shocks to the Norwegian macroeconomic system", 22 March 2022. We would like to thank participants for comments and discussion. The numerical results reported was obtained by the use of the Oxmetrics and Eviews software packages, https://www.doornik.com/products.html and https://www.eviews.com/home.html.

Appendix

A.1. Data sources and data set with CovidMod real-time forecasts and forecast errors

A.1.1. Data sources for the CovidMod project

For the real-time forecasts, a new data set was constructed as part of the preparation of each forecast. During the forecasting period documented in the paper, from 19 March 2021 to 1 April 2022, 264 data sets were constructed.

New cases, hospital admissions and hospital beds.

The sources were the internet pages of Norwegian Institute of Public Health, for new cases and hospital admissions and Norwegian Directorate of Health for hospital beds (number of persons in hospital with COVID-19).

The forecasted time series of new cases was by specimen collection date (i.e., the day of testing). This meant that when a forecast was prepared e.g., for June 5 2021 the number of cases for 4 June was preliminary and typically much lower than the final (confirmed) number of new cases (available with 3–4 day delay). To avoid having poor initialization values of new cases, the Y variable for 2–4 June was therefore nowcasted using data for new cases by date of registration. This data was obtained from the newspaper VG's COVID-19 statistics (https://www.vg.no/spesial/corona/).

Vaccination

Data for total number of persons who had received one, two and three vaccination was obtained from Norwegian Institute of Public Health web-pages.

A.1.2. Files with CovidMod forecasts and forecast errors

During the forecasting period, forecasts were published on https://normetrics.no.

Datafile with complete set of forecasts for new cases and hospital beds:
CovidMod forecasts by date.xslx.

Variable names:

NS_ddmmyy: Forecasted new cases with date of forecast origin ddmmyy (e.g., 030122 for the forecast with date of origin 3 January 2022).

INL_ddmmyy: Forecasted hospital beds with date of forecast origin ddmmyy.

NS: Actual number of new cases

INL: Actual number of hospital beds.

Time series with forecast errors (by date of origin of the forecasts) are calculated as:

$$NS_ER_ddmmyy = NS - NS_ddmmyy$$

and

$$INL_ER_ddmmyy = INL - INL_ddmmyy$$

Data files with real-time forecast errors ordered by forecast horizon (H1 to H21):
Forecast error by horizon NS.xslx (new cases)
Forecast error by horizon INL.xslx (hospital beds). The data in these two files was used in the calculation of the CovidMod RMSFEs in Table 1.

A.1.3. Sources of NIPH and Cardt forecasts

NIPH: Dated *Situational awareness and forecasting for Norway* reports. https://www.fhi.no/en/id/corona/coronavirus/coronavirus-modelling-at-the-niph-fhi/

Cardt: Forecasts of confirmed cases for Norway was downloaded from https://www.doornik.com/COVID-19/.

References

Bårdsen, G., Kolsrud, D., & Nymoen, R. (2017). Forecast robustness in macroeconometric models. *Journal of Forecasting*, *36*, 629–639. http://dx.doi.org/10.1002/for.2459.

Castle, J. L., Doornik, J. A., & Hendry, D. F. (2021a). Forecasting principles from experience with forecasting competitions. *Forecasting*, *3*(2019-w01), 138–165.

Castle, J. L., Doornik, J. A., & Hendry, D. F. (2021b). The value of robust statistical forecasts in the Covid-19 pandemic. *National Institute Economic Review*, *256*, 19–43. http://dx.doi.org/10.1017/nie.2021.9.

Clements, M. P., & Hendry, D. F. (1999). *Forecasting non-stationary economic time series*. Cambridge, MA: The MIT Press.

Clements, M. P., & Hendry, D. F. (2002). Explaining forecast failures in economics. In M. P. Clements, & D. F. Hendry (Eds.), *A companion to economic forecasting* (pp. 539–573). Oxford: Blackwell Publishers.

Doornik, J. A., Castle, J. L., & Hendry, D. F. (2022). Short-term forecasting of the coronavirus pandemic. *International Journal of Forecasting*, *38*(2), 453–466.

Doornik, J. A., Hendry, D. F., & Castle, J. L. (2020). Statistical short-term forecasting of the COVID-19 pandemic. *Journal of Clinical Immunology and Immunotherapy*, (6: 046).

Eitrheim, Ø., Jansen, E. S., & Nymoen, R. (2002). Progress from forecast failure: The Norwegian consumption function. *The Econometrics Journal*, *5*, 40–64.

Falch, N. S., & Nymoen, R. (2011). The accuracy of a forecast targeting central bank. *Economics: The Open-Access, Open-Assessment E-Journal*, *5*(15), 1–36.

Fayomi, A., Nasir, J. A., Algarni, A., Rasool, M. S., Jamal, F., & Chesneau, C. (2022). Best selected forecasting models for COVID-19 pandemic. *Open Physics*, *20*(1), 1303–1312.

Granger, C. W. J. (1973). On the properties of forecasts used in optimal policy decisions. *Journal fo Public Economics*, *2*, 347–356.

Granger, C. W. J., & Newbold, P. (1986). Forecasting economic time series. San Diego: Academic Press.

Harvey, A. C. (1993). *Time series models* (2nd ed.). Hempstead: Harvester Wheatsheaf.

Harvey, A. C., & Kattuman, P. (2021). A farewell to R: Time series models for tracking and forecasting epidemics. *Journal of the Royal Society Interface*, (18), 20210179. http://dx.doi.org/10.1098/rsif.2021.0179.

Hyndman, R. J. (2023). Forecasting, causality and feedback. *International Journal of Forecasting*, *39*, 558–560.

Johansen, L. (1972). On the optimal use of forecasts in economic policy decisions. *Journal of Public Economics*, *1*, 1–25.

Korolev, I. (2021). Identification and estimation of the SEIRD epidemic model for COVID-19. *Journal of Econometrics*, *220*(1), 63–85.

Li, M. L., Bouardi, H. T., Lami, O. S., Trikalinos, T. A., Trichakis, N., & Bertsimas, D. (2023). Forecasting COVID-19 and analyzing the effect of government interventions. *Operations Research*, *71*(1), 184–201.

Li, S., & Linton, O. (2021). When will the Covid-19 pandemic peak? *Journal of Econmetrics*, *220*(1), 130–157.

Mills, T. C. (2022). Modelling the link between Covid-19 cases, hospital admissions and deaths in England. *National Accounting Review*, *4*(1), 38–58. http://dx.doi.org/10.3934/NAR.2022003.

Nymoen, R. (2019). *Dynamic econometrics for empirical macroeconomic modelling*. Boston: World Scientific.

Osborn, D. R. (2002). Unit-root versus determinstic representations of seasonality for forecasting. In M. P. Clements, & D. F. Hendry (Eds.), *A companion to economic forecasting* (pp. 409–430). Blackwell Publishers.

Petropoulos, F., & Makridakis, S. (2020). Forecasting the new coronavirus COVID-19. *Plos ONE*, *15*, Article e0231236.

Teräsvirta, T., Tjøstheim, D., & Granger, C. W. J. (2010). *Modelling nonlinear economic time series*. Oxford University Press.

van Dijk, D., Teräsvirta, T., & Franses, P. H. (2002). Smooth transition autoregressive models — A survey of recent developments. *Econometric Reviews*, *21*(1), 1–47.